

Modbus-RTU Communication Protocol for Single-Phase Multi-Rate Prepaid Electricity Meters (SW Universal Version)

I. Basic Agreement Specifications

1.1 Physical Layer Parameters (default Configurable)

Parameter	Span	Windows default
Baud rate	1200/2400/4800/9600 bps	9600 bps
Data bit	8 bit	8 bit
Stop bit	1 bit /2 bit	1 bit
Check bit	No parity/odd parity/even parity	No validation
Communication medium	RS-485 shielded twisted pair (A/B terminals)	-
Slave address	1~247(0 is the broadcast address, no response)	1

1.2 Frame Structure (Modbus-RTU Standard)

Slave address (1 byte) → Function code (1 byte) → Data field (N bytes) → CRC check (2 bytes,0xA001 polynomial, low byte first)

- The frame-to-frame byte interval is ≤ 1.5 character time, and the frame-to-frame interval is ≥ 3.5 character time.

1.3 Data Format Specification

Data type	Byte length	Applicable scene	Calculation factor/Note
UINT16	2	Voltage/Current/Frequency/Status Word	-

INT16	2	Power factor	PF: 0.001
INT32	4	Active power per phase / reactive power per phase / total power	Power: 0.0001 (kW/kvar); Range: -2147.4836 to 2147.4835 kW
UINT32	4	Forward/reverse power/demand	0.01 kWh (electricity); 0.001 kW (demand)
INT32	4	Amount type (Balance/Recharge/Refund)	0.01 Yuan
UINT8×N	N	Time Zone Table/Interval Table/Table Number	BCD code / binary code

2. Core Register Definitions (By Function)

2.1 Regular Data Acquisition Register (read-Only, 0000H~00FFH)

Register address (H)	Data name	Data type	Length (bytes)	Remarks
0000	Total active power	UINT32	4	0.01 kWh , the total of peak, flat, and valley rates 5-8
0002	Current active peak power	UINT32	4	0.01 kWh (Rate 1)
0004	Current active peak power	UINT32	4	0.01 kWh (Rate 2)
0006	Current active power	UINT32	4	0.01 kWh (Rate 3)
0008	Active valley power	UINT32	4	0.01 kWh (Rate 4)

000A	Current active power rate 5 electric energy	UINT32	4	0.01 kWh
000C	Current active power rate 6 electric energy	UINT32	4	0.01 kWh
000E	Current active power rate 7 electric energy	UINT32	4	0.01 kWh
0010	Current active power rate 8 electric energy	UINT32	4	0.01 kWh
0012	Total active power	UINT32	4	0.01 kWh
0014	Reverse total active power and total electric energy	UINT32	4	0.01 kWh
0016	Voltage	UINT16	2	0.1 V (three-phase meter)
0017	Current	UINT16	2	0.01 A (three-phase meter)
0018	Active power	INT16	4	0.0001 kW , positive and negative indicate direction
001A	Reactive power	INT16	4	0.0001 kvar , positive and negative indicate direction
001C	Apparent output	UINT16	4	0.0001 kVA
001E	Power factor	INT16	4	0.001 (- 1000~1000)
001F	Grid frequency	UINT16	2	0.01 Hz
0020	Date and time (year/month)	UINT8×2	2	BCD code, format: year (2 bits) +

				Month(2 digits)
0021	Date and time (day and hour)	UINT8×2	2	BCD code, format: day (2 bits) + hour (2 bits)
0022	Date and time (minutes and seconds)	UINT8×2	2	BCD code, format: minute (2 bits) + second (2 bits)
0023	Running state word	UINT16	2	0x0001 : Close; 0x0002 : Trip; 0x0004 : Low balance alarm

2.2 Core Register for Recharge/Refund (Read/Write, 00A0H~00B2H)

Register address (H)	Data name	Data type	Length (bytes)	Read-write	Remarks
00A0	Reported amount 1(low balance)	INT32	4	R/W	0.01 If the value is lower than this threshold, an alarm will be triggered.
00A2	Report amount 2(overdue)	INT32	4	R/W	0.01 If the value is lower than this threshold, the circuit breaker will trip.
00A4	Limit of credit	UINT32	4	R/W	0.01 Yuan, maximum allowable credit amount
00A6	Single top-up amount	INT32	4	W	0.01 Write a positive number in yuan Confirm to take effect

00A8	Balance	INT32	4	R	0.01 Yuan, Recharge Accumulation/ Refund Deduction
00AA	Single refund amount	INT32	4	W	0.01 yuan, positive number, balance after deduction ≥ 0
00AC	Total electricity purchase amount	INT32	4	R	0.01 Total Recharges Forehead (cannot be cleared)
00AE	Total refund amount	INT32	4	R	0.01 Total refund amount Forehead (cannot be cleared)
00B0	Number of electricity purchases	UINT16	2	R	0~1000 cumulative top-up attempts
00B1	Operation status word	UINT16	2	R	0x0001: Recharge successful; 0x0002: Refund successful; 0x0003: Operation failed
00B2	Recharge/ Refund Confirmation	UINT16	2	W	0x0001: Confirm; 0x0000: Cancel (prevent accidental operation)

2.3 Rate and Tariff Configuration Register (read/write, 5038H~5047H)

Register address (H)	Data name	Data type	Length (bytes)	Read-write	Remarks
00A0	Reported amount 1(low balance)	INT32	4	R/W	0.01 If the value is lower than this threshold, an alarm will be triggered.
00A2	Report amount 2(overdue)	INT32	4	R/W	0.01 If the value is lower than this threshold, the circuit breaker will trip.
00A4	Limit of credit	UINT32	4	R/W	0.01 Yuan, maximum allowable credit amount
00A6	Single top-up amount	INT32	4	W	0.01 Write a positive number in yuan Confirm to take effect
00A8	Balance	INT32	4	R	0.01 Yuan, Recharge Accumulation/ Refund Deduction
00AA	Single refund amount	INT32	4	W	0.01 yuan, positive number, balance after deduction ≥ 0
00AC	Total electricity purchase amount	INT32	4	R	0.01 Total Recharges Forehead (cannot be cleared)
00AE	Total refund amount	INT32	4	R	0.01 Total refund amount Forehead (cannot be cleared)

00B0	Number of electricity purchases	UINT16	2	R	0~1000 cumulative top-up attempts
00B1	Operation status word	UINT16	2	R	0x0001: Recharge successful; 0x0002: Re-fund successful; 0x0003: Operation failed
00B2	Recharge/Refund Confirmation	UINT16	2	W	0x0001: Confirm; 0x0000: Cancel (prevent accidental operation)
00C1	Forced control word	UINT16	2	RW	0x0001: Strong Control open 0x0000: Force closed
00C2	Split control word	UINT16	2	RW	0x0000: Strong 0x0001: Forced trip

Time Zone Table/Interval

Register address (H)	Data name	Data type	Length (bytes)	Read-write	Remarks
5038~5039	T1 (peak) tariff	UINT32	4	R/W	0.0001 ¥/kWh (Rate 1)
503A~503 B	T2 (peak) electricity price	UINT32	4	R/W	0.0001 Yuan/kWh (cost)

					Rate 2)
503C~503D	T3 (flat) electricity price	UINT32	4	R/W	0.0001 yuan/kWh (tariff 3)
503E~503 F	T4 (valley) tariff	UINT32	4	R/W	0.0001 yuan/kWh (tariff 4)
5040~5041	T5 Electricity price	UINT32	4	R/W	0.0001 yuan/kWh (tariff 5)
5042~5043	T6 Electricity price	UINT32	4	R/W	0.0001 Yuan/kWh (Rate 6)
5044~5045	T7 Electricity price	UINT32	4	R/W	0.0001 yuan/kWh (tariff 7)
5046~5047	T8 electricity price	UINT32	4	R/W	0.0001 yuan/kWh (tariff 8)

2.4 Time Zone/Interval Configuration Register (Read/Write, 2000H~20D1H)

2.4.1 Time Zone Table (14 Time Zones, 2000H~2014H,42 Bytes)

Register address (H)	Data name	Data format (3 bits per hour zone)	Remarks
2000~2014	14 Time Zone Configuration	Schedule number (1 byte) + Start date (1 byte) + Start month (1 byte)	Each time zone occupies 3 bytes, and 14 time zones occupy 42 bytes. Time slot number: 1~8(corresponding to 8 rates)

Instance	Time Zone 1	01 01 01	Schedule 1, effective on 1 1
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2.4.2 Schedule Table (8 Tables, 202AH~20D1H, Each 42 Bytes)

Each schedule supports 7 time slots, each occupying 6 bytes. Format:

Rate number (1 byte) + Start time (1 byte) + Start minute (1 byte) + End time (1 byte) + End minute (1 byte) + Standby (1 byte)

Register address (H)	Schedule Table Name	Overlapping bytes	Remarks
202A~203E	Schedule 1	42	Applicable rate 1-8, time period: 00:00~23:59
203F~2053	Schedule 2	42	-
2054~2068	Schedule 3	42	-
2069~207D	Schedule 4	42	-
207E~2092	Schedule 5	42	-
2093~20A7	Schedule 6	42	-
20A8~20BC	Schedule 7	42	-
20BD~20D1	Schedule 8	42	-
Instance	Schedule 1-Period 1	01 08 00 11 00 00	Rate 1(sharp), 8:00~11:00

2.5 Communication Parameter Configuration Register (read/write, 0330H~0331H)

Register address (H)	Data name	Data format (UINT8×2)	Read-write	Remarks
0330	Address/Port Rate	Address (1)ports	R/W	Address : 1~247; Porter

		Rate(1)		Rate:0=1200/ 1=2400/2= 4800/3=9600
0331	Check stop bit	Check bit (1) Stop bit (1)	R/W	Check bit:0= None /1=Odd / 2=Even; Stop bit:0=1 bit /1=2 bit

3. Core Operation Command Example (Slave Address=01H)

3.1 Read 8 Tariff Electricity Data (0000H~0010H)

- Master station request:01 03 00 00 00 08 44 09
.01: Slave address;03: Read register;00 00: Starting address;00 08: Read 8 registers (16 bytes, covering 4 frequency power);44 09:CRC
- Reply from the station:01 03 10 00 00 07 D0 00 00 03 E8... 79 8B
.10: Data length (16 bytes);00 00 07 D0=2000 → Total active power 20.00 kWh;00 00 03 E8=1000 → Peak power: 10.00 kWh

3.2 Set the T1 (peak) Tariff (5038H~5039H,0.50 Yuan/kWh → Register Value 5000)

- Master station request:01 10 50 38 00 02 04 00 00 13 88 B8 3A
.10: Write multiple registers;50 38: Starting address;00 02:2 registers;04: Data length;00 00 13 88=5000;B8 3A:CRC
- Response from the station:01 10 50 38 00 02 41 C8(Write successful)

3.3 Configure Schedule 1 (202AH, 8:00~11:00, Rate 1)

- Master station request:01 10 20 2A 00 01 06 01 08 00 11 00 00 78 0A
.20 2A: starting address;00 01:1 register (6 bytes);06: data length;01 08 00 11 00 00: Time slot configuration; 78 0A: CRC
- Response from the station:01 10 20 2A 00 01 40 09(configuration confirmed)

3.4 Recharge 50 Yuan(00A6H Writes 5000 + 00B2H Confirms)

1. Write recharge amount:01 10 00 A6 00 02 04 00 00 13 88 74 AB→ Response:01 10 00 A6 00 02 A1 EB
2. Confirm recharge:01 10 00 B2 00 01 02 00 01 7C 42→ Response:01 10 00 B2 00 01 98 0B D3 DB
3. Query status:01 03 00 B1 00 01 D4 2D → Response:01 03 02 00 01 79 8B(00 01= Recharge successful)

4. Exception Handling Notes

4.1 Common Error Codes (function Code + 0x80)

Exception code	Meaning	Search Direction
01	Illegal function code	Confirm the function code is 03/06/10(read/write register)
02	Illegitimate register address	Check if the register address is within the device's supported range
03	Invalid data value	Does the amount or electricity price exceed the device limit (e.g., refund>remaining amount)
04	Execution failed	The operation requires two-step confirmation (if the recharge is not confirmed)